MILESTONE 10 - AGENT ARCHITECTURE

Architecture Pattern
Implemented a planner-executor-reviewer architecture in backend/agent_architecture.py that extends the current backend without replacing existing endpoints.

1. **Planner Agent**
   - Converts user requests into explicit step-by-step plans.
   - Identifies actions such as budget updates, expense parsing, expense writes, query, and budget comparison.
   - Produces ordered steps with dependency references for tool outputs.

2. **Executor Agent**
   - Resolves step dependencies and runs tools in order.
   - Calls existing backend capabilities as tools:
     - parse_expense_text → llm_pipeline.parse_expense
     - add_expense → db.create_expense
     - set_budget → db.create_budget
     - query_expenses → db.get_expenses
     - get_budget_comparison → db.get_budget_comparison
     - get_cost_of_living → col_api.get_city_data
   - Captures per-step execution records with input, output, status, and errors.

3. **Reviewer Agent**
   - Evaluates execution results.
   - Returns final task status (completed or partial) and synthesized response.
   - Preserves plan and execution trace for auditability.

Memory and State Management
- Added InMemoryAgentStateStore and AgentSessionState.
- Session state persists:
  - turn_count
  - recent_tasks
  - last_plan
  - last_step_outputs
  - working_memory
- Supports multi-turn continuation using session_id.

API Endpoints Added

- POST /api/agent/execute
    - Authenticated endpoint for multi-step agent task execution.
    - Request supports task, optional session_id, and optional context.
- GET /api/agent/session/{session_id}
- Authenticated endpoint to inspect persisted session state.

Multi-Step Test Coverage

Added tests/test_agent_architecture.py (async unittest):

1. **Budget + expense + remaining flow**
- Verifies plan generation and execution for a combined multi-step request.
- Confirms budget write + expense write + budget comparison completion.

2. **Session memory continuity**
- Runs two turns with the same session_id.
- Verifies turn count and recent task persistence.

3. **Cost-of-living + query flow**
- Validates multi-step reasoning that combines external context with expense querying.